Chapter 2 (Bluman)

Frequency Distributions and Graphs

Data Representation

Data presentation is the method of organizing, summarizing, and showing information using tools such as tables, charts, diagrams, histograms, and graphs.

Objectives of Data Presentation

1. **Simplify raw data:** Makes large amounts of raw data easier to understand and analyze.
2. **Remove complexity:** Removes unnecessary details and highlights the important features of the data.
3. **Make comparison easier:** Helps us compare different sets or groups of data and draw conclusions.
4. **Help in statistical analysis:** Divides data into similar groups so we can easily identify similarities and differences.

Frequency Distribution

Frequency distribution is a table that organizes a data set into a suitable number of classes or groups and shows the frequency of each class.

To Construct a Frequency Distribution of Qualitative Data

Step 1 List the distinct values of the observations in the data set in the first column of a table.

Step 2 For each observation, place a tally mark in the second column of the table in the row of the appropriate distinct value.

Step 3 Count the tallies for each distinct value and record the totals in the third column of the table.

Types of Frequency Distribution

1. Ungrouped Frequency Distribution

Data is shown using **individual values**, and the frequency of each value is given.

Example:

Marks	Frequency
10	2
20	3
30	4

2. Grouped Frequency Distribution

Data is divided into **classes or intervals**, and the frequency of each class is given.

Example:

Marks	Frequency
0–10	2
11–20	5
21–30	7

3. Percentage Frequency

Percentage frequency shows the frequency of a value or class as a percentage of the total frequency.

Formula

$$\text{Percentage Frequency} = \frac{\text{Frequency}}{\text{Total Frequency}} \times 100$$

Example:

If 20 students are in total and 5 students scored 80:

$$\frac{5}{20} \times 100 = 25\%$$

So, the **percentage frequency is 25%**.

4. Cumulative Frequency Distribution

Cumulative frequency distribution is a table that shows the running total of frequencies up to each class.

In simple words, we **keep adding the frequencies** as we move from one class to the next.

Example:

Marks	Frequency	Cumulative Frequency
0–20	5	5
21–40	8	13
41–60	12	25
61–80	10	35

Here:

- First: **5**
- Then: **5 + 8 = 13**
- Then: **13 + 12 = 25**
- Then: **25 + 10 = 35**

So, the last cumulative frequency is always equal to the **total frequency**.

5. Relative Cumulative Frequency (RCF)

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Relative Cumulative Frequency (RCF) shows the **cumulative frequency compared to the total frequency**.

Formula:

$$RCF = \frac{\text{Cumulative Frequency}}{\text{Total Frequency}}$$

If expressed as a percentage:

$$RCF = \frac{\text{Cumulative Frequency}}{\text{Total Frequency}} \times 100$$

Example:

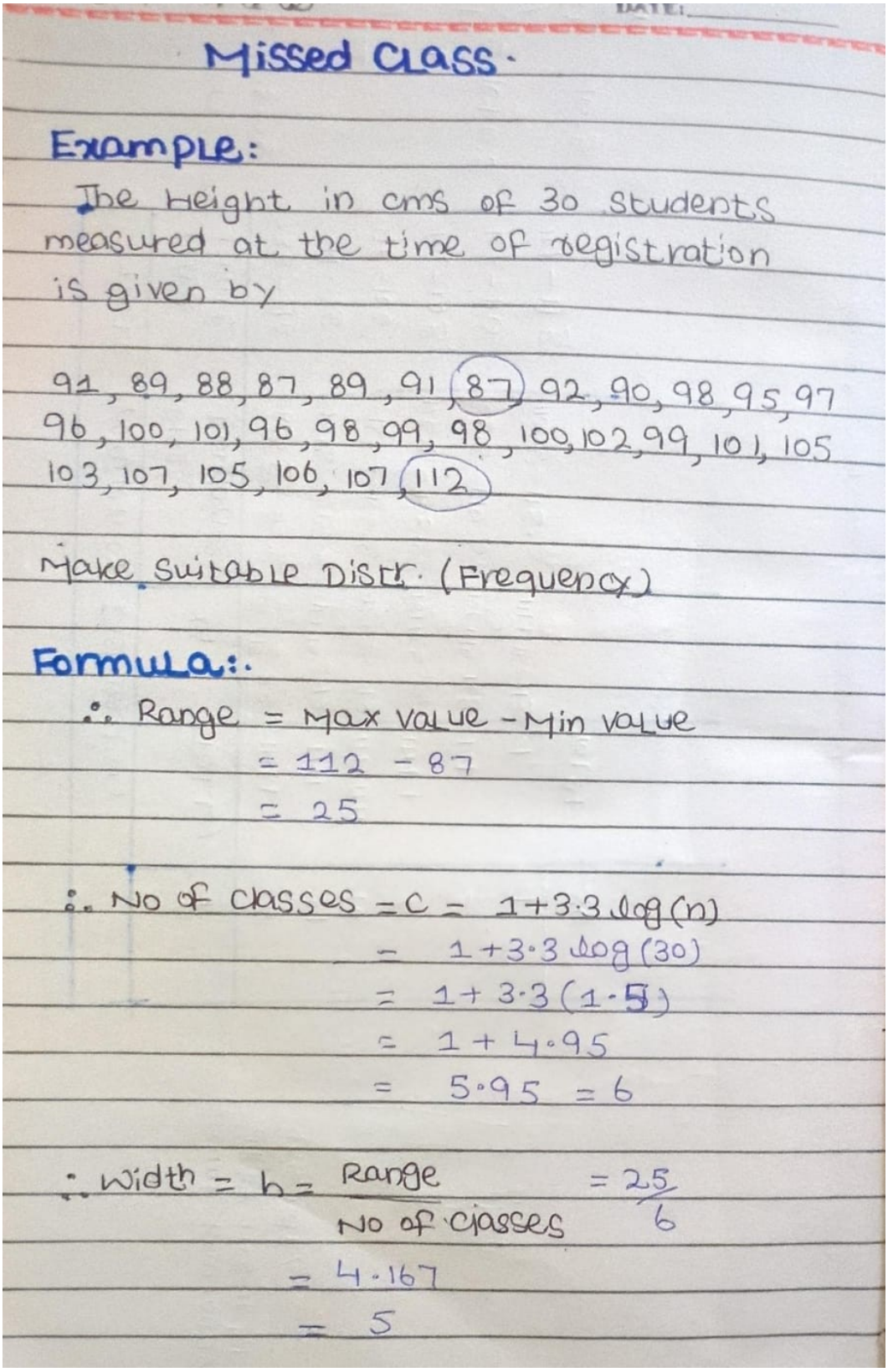
Cumulative frequency = 15

Total frequency = 30

$$RCF = \frac{15}{30} = 0.5$$

or **50%**.

Example



Missed Class.

Example:

The Height in cms of 30 students measured at the time of registration is given by

91, 89, 88, 87, 89, 91, 87, 92, 90, 98, 95, 97
96, 100, 101, 96, 98, 99, 98, 100, 102, 99, 101, 105
103, 107, 105, 106, 107, 112

Make Suitable Distr. (Frequency)

Formula:

∴ Range = Max value - Min value
= 112 - 87
= 25

∴ No of classes = C = 1 + 3.3 log(n)
= 1 + 3.3 log(30)
= 1 + 3.3 (1.5)
= 1 + 4.95
= 5.95 = 6

∴ width = h = Range / No of classes = 25/6
= 4.167
= 5

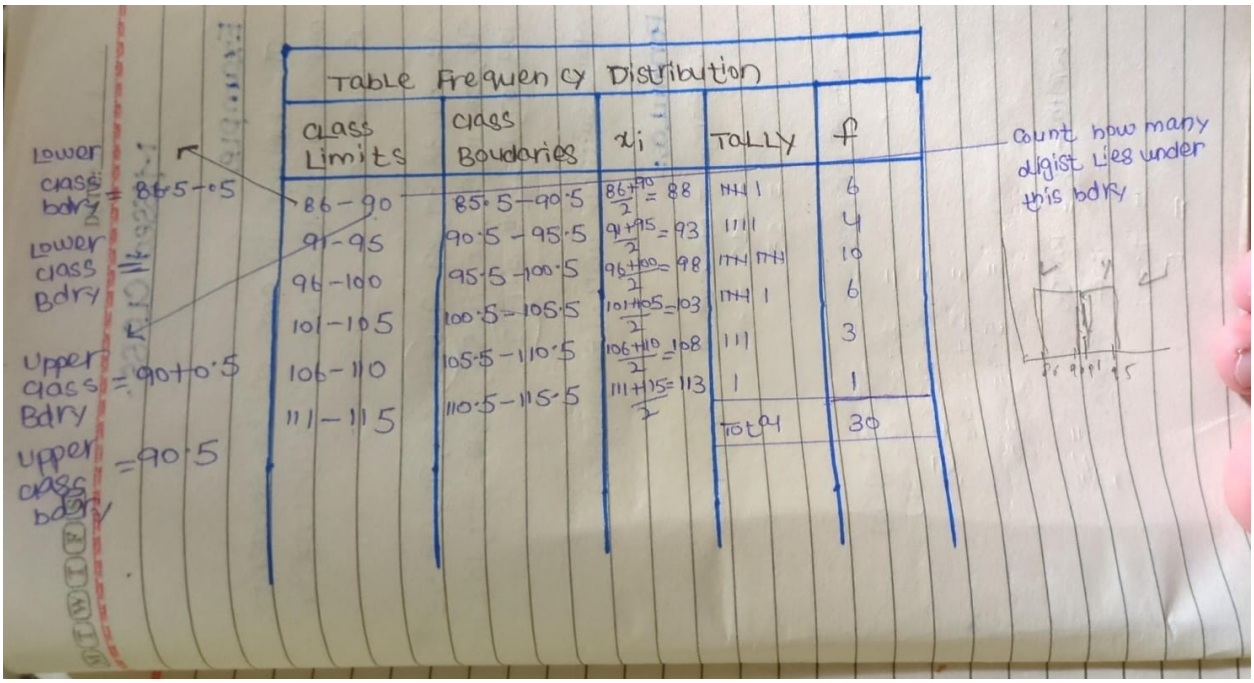


Table Frequency Distribution

Class Limits	Class Boudaries	x_i	TALLY	f
86 – 90	85.5 – 90.5	$\frac{86+90}{2} = 88$	𝍸 I	6
91 – 95	90.5 – 95.5	$\frac{91+95}{2} = 93$	IIII	4
96 – 100	95.5 – 100.5	$\frac{96+100}{2} = 98$	𝍸 𝍸	10
101 – 105	100.5 – 105.5	$\frac{101+105}{2} = 103$	𝍸 I	6
106 – 110	105.5 – 110.5	$\frac{106+110}{2} = 108$	III	3
111 – 115	110.5 – 115.5	$\frac{111+115}{2} = 113$	I	1
			Total	30

Lower class bdry = 86 - 0.5

Lower class Bdry

Upper class Bdry = 90 + 0.5

Upper class bdry = 90.5

count how many digist lies under this bdry

Relative Freq:

$$\therefore \text{R.f} = \frac{\text{observed Freq}}{\text{Total Freq}}$$

Commulative Freq:

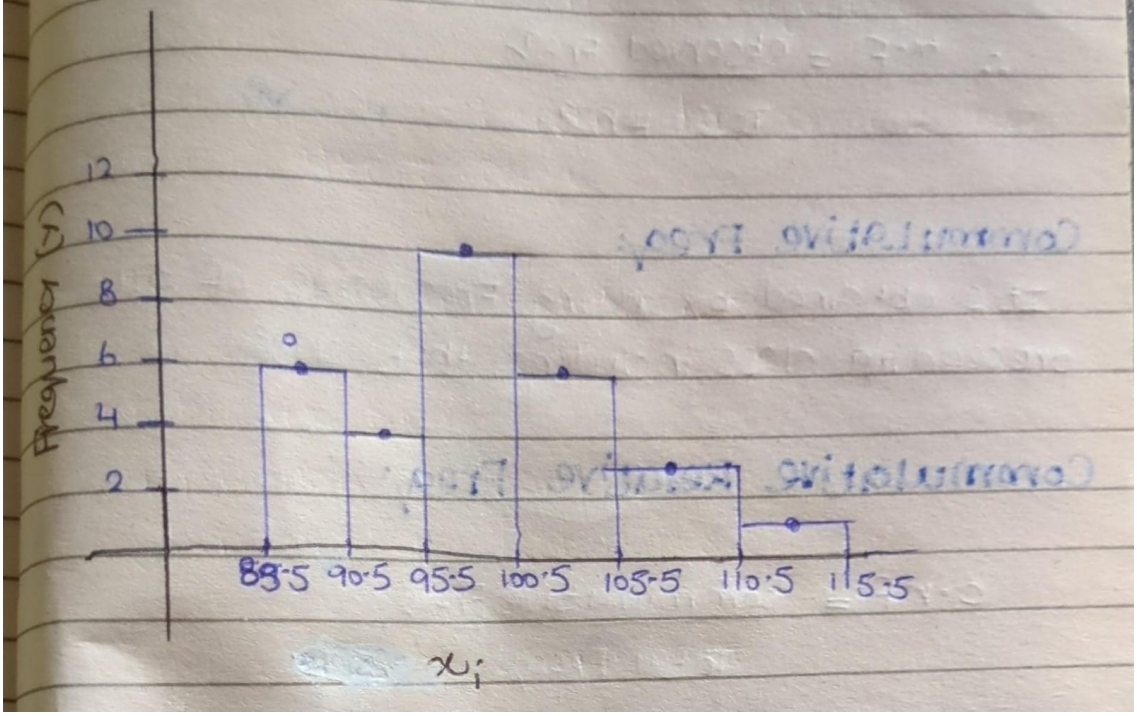
It's obtained by adding Frequecies of all preceeding class including that class

Commulative Relative Freq:

$$\text{C.R.f} = \frac{\text{commulative Freq}}{\text{Total Freq}}$$

class boudaries	f	R.f	C.f	C.R.f
85.5-90.5	6	$6/30 = 0.2$	6	$6/30$
90.5-95.5	4	$4/30 = 0.1$	$6+4=10$	$10/30$
95.5-100.5	10	$10/30 = 0.3$	$10+10=20$	$20/30$
100.5-105.5	6	$6/30 = 0.2$	$20+6=26$	$26/30$
105.5-110.5	3	$3/30 = 0.1$	$26+3=29$	$29/30$
110.5-115.5	1	$1/30 = 0.03$	$29+1=30$	$30/30$
	30			

Histrogram



Frequency Polygon | Curve

